Assignment Four
Visual Question Answering

Omar Walied 7058 , Hend Khaled 6986 , Habiba Hefny 6939

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1 Introduction

VQA is a rapidly advancing field that merges computer vision and natural language processing to enable machines to understand and answer questions about images. Its applications range from robotics to autonomous vehicles, virtual assistants, and augmented reality, promising a future where machines can comprehend and interact with visual information in a human-like manner.

2 Problem Description

Our focus is on VizWiz dataset, which is designed for training models to assist visually impaired individuals. The dataset includes images and spoken questions, providing a real-world context for visually impaired people, aiming to improve their understanding of visual information by using VQA techniques. VizWiz-VQA contains diverse answers crowd-sourced from the community. We analyze the dataset's characteristics and explore VQA methodologies to enhance visual perception for visually impaired individuals. By training models on this dataset, we aim to improve accessibility and inclusivity, ultimately enhancing the quality of life for visually impaired individuals.

3 Code Explaination

3.1 Notebook Setup and Dependencies

- Creating a new Kaggle notebook
- Adding the dataset and installing necessary dependencies and libraries
- Importing libraries such as PyTorch, torchvision, numpy, cv2, matplotlib, etc.

3.2 Device Configuration

- Setting the device variable based on the availability of a CUDA-enabled GPU
- Choosing between "cuda" and "cpu" for computations

3.3 Reading and Processing JSON Data

- Reading the contents of a JSON file and storing it as a string
- Converting the string into an object based on the JSON structure
- Processing the data and generating two sets of labels (labels1 and labels2)
- Plotting the frequency of each letter in the label string

3.4 Custom Dataset Class

- Defining a custom dataset class named "myDataset"
- Inheriting from the torch.utils.data.Dataset class
- Implementing methods: init, getitem, and len
- Initializing the dataset and storing data, labels, and a train flag
- Retrieving an item from the dataset based on the index
- Returning the length of the dataset

3.5 Neural Network Model

- Building a neural network model inherited from torch.nn.Module class
- Defining layers such as linear, activation, normalization, dropout, and attention
- The model architecture processes input data and generates predictions for two labels
- Initializing the model in the init method and implementing the forward pass in the forward method

3.6 Data Splitting and Dataloaders

- Splitting the data into training, validation, and testing sets
- Creating dataloaders to efficiently load the data during training, validation, and testing

3.7 Model Training and Evaluation

- Defining the run model function for the training and evaluation loop
- Running the model on a specified dataloader and updating the parameters with an optimizer
- Recording training and validation loss and accuracy values for each epoch
- Visualizing the training and validation loss, and training and validation accuracy

3.8 Model Evaluation on Test Data

- Evaluating the performance of the trained model on the test dataset
- Calculating loss and accuracy metrics

3.9 Continue of Evaluation (Answerability)

- Define a new custom dataset class. It is designed for evaluating the model's answerability.
- we perform the previous steps (3.4, 3.5, 3.6, 3.7, 3.8)

4 Conclusion

This report focuses on addressing the challenges faced by visually impaired individuals through visual question answering (VQA) techniques applied to the VizWiz dataset. Custom dataset and model classes were developed to handle the dataset and perform VQA tasks, incorporating various techniques such as linear transformations, normalization, dropout, and attention mechanisms. Data loaders were created to efficiently handle data during training, validation, and testing. The training process involved running the model on the data loader, collecting metrics, and visualizing the progress. The trained model was evaluated on a test dataset, aiming to improve the accessibility and comprehension of visual information for visually impaired individuals. The approach aimed to combine computer vision and natural language processing to enhance their quality of life.